

Assignment Title: Python Data Handling with Pandas and NumPy

Objective

The aim of this assignment is to enhance your Python programming skills and to apply Pandas and NumPy for data manipulation and analysis. You will work with Python basics, handle data using Pandas, and perform numerical operations with NumPy.

Instructions

1. Python Programming Basics

- **Task 1: Python Functions and Lists**

- **Objective:** Create and use functions and lists in Python.
- **Instructions:**
 1. Write a Python function that takes a list of numbers and returns a new list with each number squared.
 2. Create a list of 10 integers, use your function to square these numbers, and print the original and squared lists.

- **Task 2: Dictionary Operations and Comprehensions**

- **Objective:** Work with dictionaries and dictionary comprehensions in Python.
- **Instructions:**
 1. Create a dictionary where the keys are names of students and the values are their scores in an exam.
 2. Write a function that takes this dictionary and returns a new dictionary with only those students who scored above 50.
 3. Use a dictionary comprehension to create a dictionary where the keys are student names and the values are their scores multiplied by 1.1 (e.g., adding 10% to the score).

2. Data Manipulation with Pandas

- **Task 1: Creating and Exploring a DataFrame**

- **Objective:** Create a Pandas DataFrame from a dictionary and explore its contents.
- **Instructions:**

Use the following dictionary to create a Pandas DataFrame:

```
data = {  
    'Name': ['Alice', 'Bob', 'Charlie', 'David', 'Eva'],  
    'Age': [25, 30, 35, 40, 45],  
    'City': ['New York', 'Los Angeles', 'Chicago', 'Houston', 'Phoenix']  
}
```

1. Create a DataFrame from this dictionary.
2. Display the first few rows of the DataFrame.
3. Calculate and print the average age from the DataFrame.

- **Task 2: Data Filtering and Aggregation**

- **Objective:** Filter and aggregate data in a DataFrame.
- **Instructions:**
 1. Filter the DataFrame to include only rows where the age is greater than 30.
 2. Add a new column to the DataFrame called **AgeGroup**, where the value is '**Senior**' if age is greater than or equal to 40, otherwise '**Adult**'.
 3. Group the data by **AgeGroup** and calculate the average age for each group.

3. Data Manipulation with NumPy

- **Task 1: NumPy Array Operations**

- **Objective:** Perform operations on NumPy arrays.
- **Instructions:**
 1. Create a NumPy array with 12 random integers between 1 and 50.
 2. Reshape the array into a 3x4 matrix.
 3. Find and print the maximum, minimum, and mean values of the array.

- **Task 2: Array Indexing and Slicing**

- **Objective:** Practice indexing and slicing with NumPy arrays.
- **Instructions:**

1. Create a NumPy array of shape (4, 3) with values from 1 to 12.
 2. Extract and print the second row of the array.
 3. Extract and print a 2x2 submatrix from the top-left corner of the array.
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Deliverables

1. **Python Script or Jupyter Notebook:** Include code for:
 - Python programming tasks (functions and lists, dictionary operations).
 - Pandas tasks (DataFrame creation, exploration, filtering, and aggregation).
 - NumPy tasks (array operations, indexing, and slicing).